THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A concave lens is also called a: 1

(a) diverging lens **(b)** converging lens **(c)** magnifier **(d)** plane lens

2. In a plane mirror, the image of an object is: 1

(a) smaller than the object **(b)** larger than the object
(c) inverted **(d)** of the same size as the object

3. A mirror that bulges outwards is called: 1

(a) concave mirror **(b)** convex mirror
(c) cylindrical mirror **(d)** plane mirror

4. A magnifying glass is a: 1

(a) convex mirror **(b)** plane mirror **(c)** convex lens **(d)** concave lens

5. A boy stands 2 m in front of a plane mirror. The distance between him and his image is: 1

(a) 8 m **(b)** 1 m **(c)** 2 m **(d)** 4 m

6. The point at which parallel rays meet after reflection from a concave mirror is its: 1

(a) pole **(b)** centre of curvature
(c) aperture **(d)** principal focus

7. Spectacles for a short-sighted person use: 1

(a) concave lenses **(b)** convex mirrors **(c)** convex lenses **(d)** plane glass

8. The image of the letter 'P' seen in a plane mirror appears: 1

(a) unchanged **(b)** laterally inverted
(c) smaller **(d)** upside down only

9. The distance between the pole and the principal focus is the: 1

(a) radius of curvature **(b)** object distance
(c) focal length **(d)** aperture

10. The image formed by a concave lens is always: 1

(a) real and enlarged **(b)** real and inverted
(c) virtual and enlarged **(d)** virtual and diminished

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. An object is placed 10 cm from a plane mirror. Find the distance of the image from the object. 2

12. What is a virtual image? Give an example. 2

13. State two uses of convex mirrors. 2

14. Name two devices using multiple reflections. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two plane mirrors are inclined at 60 degrees. How many images of an object are formed? Show the working. 3

16. Why do torches and headlights use concave reflectors? 3

17. Why does a plane mirror reverse left and right but not up and down? 3

OR

Explain the working of a periscope. 3

18. State three differences between a convex and a concave lens. 3

OR

Differentiate between a real and a virtual image (three points). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State the rules for ray diagrams by concave mirrors and describe the nature of image when the object is placed beyond C, between C and F, and between F and P. 5

OR

Explain the uses of mirrors and lenses in optical instruments such as the periscope, camera, microscope and telescope. 5

20. Compare the uses of plane, concave and convex mirrors with two examples each. 5

OR

Explain how regular and diffuse reflection explain why we see a mirror's image but also objects from all directions. 5

END OF QUESTION PAPER — Total: 40 Marks

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